

- 41) Evaluate the distance (d) between two points $A(1, -3)$ and $B(1, 5)$ (2 marks)
- 42) If $A = (0.2, 0.5, 0.3)$ and $B = (0.3, 0.6, 0.1)$ are two PFs, justify that $A \leq B$. (2 marks)
- 43) Define a fuzzy set (2 marks)
- 44) Let x be a fuzzy number and Δ be an idempotent t -norm then, find the value of $(x \Delta x) \Delta x$? (2 marks)
- 45) Let $X = \{x_i\}$ be a finite set and $A = \{x_1, (0.8, 0.2)\}$, $B = \{x_1, (0.7, 0.3)\}$ are two fuzzy sets on X , evaluate $A \oplus B$. (3 marks)
- 46) Evaluate the Euclidean distance between two PFs. $A = (0.4, 0.5, 0.1)$ and $B = (0.2, 0.6, 0.2)$. (3 marks)
- 47) Define strictly monotone t -norm (2 marks)
- 48) Find the ∇ -implication, where $x \nabla y = x + y - xy$ (3 marks)
- 49) Let $X = \{x_1, x_2, x_3, x_4\}$ be a Universal set and $A = \{(x_1, 0.6), (x_2, 0.7), (x_3, 1.0), (x_4, 0.9)\}$, $B = \{(x_1, 0.6), (x_2, 0.3), (x_3, 0.0), (x_4, 0.4)\}$ are two fuzzy sets on X then show that normalized Hamming distance $d(A, B) > 0$ (5 marks)
- 50) If $A = (0.3, 0.4, 0.2)$ and $B = (0.3, 0.5, 0.1)$ are two PFs, evaluate
 i) $A \subseteq B$, ii) $A \cup B$, iii) $A \cap B$ (5 marks)
- 51) Define Alpha cut in Relations. And explain it with an example. (5 marks)

- 1) Distance between two point $A(3, -3)$ & $B(3, 5) = 8$
- 2) Distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- 3) Normalized Euclidean distance formula $= \frac{1}{n} \sqrt{\sum_{i=1}^n (m_A(x_i) - m_B(x_i))^2}$
- 4) Hamming distance formula $= \sum_{i=1}^n |m_A(x_i) - m_B(x_i)|$
- 5) max(min) b/w $A \in B$ where $A(x, 0.4)$, $B(x, 0.7) = 0.4$
- 6) $A(x, 0.6)$, $B(x, 0.3)$, $x \in \{x_i\}$ max(min) distance b/w $A \in B = 0.3$
- 7) $A(x, 0.2)$, $B(x, 0.5)$ then max(A, B) = 0.5
- 8) $A(x, 0.3)$, $B(x, 0.6)$ min(A, B) = 0.3
- 9) Universal set Property --- $\text{dist}_{\text{fuzzy}}(A, B)$ condition $0 \leq d(A, B) \leq 1$
- 10) Universal set Property $A = B$ condition
- 11) membership degree is called Picture fuzzy set
- 12) If μ is 0.6 then non-membership degree of it is 0.6
- 13) $A = B$ in set $\mu_A(x) = \mu_B(x)$, $\nu_A(x) = \nu_B(x)$
- 14) $A \subseteq B$ $\mu_A(x) \leq \mu_B(x)$, $\nu_A(x) \geq \nu_B(x)$
- 15) ~~two PFS~~ $A = B$ if $A \subseteq B$ and $B \subseteq A$
- 16) $P(0.2, 0.5, 0.3)$ PFS its complement $(P^c, 0.3, 0.5, 0.2)$ Similarity norm
- 17) $P(x \in B)$ is a PFS, then $\gamma_{A \rightarrow B} = \frac{1 - \mu_B}{1 - \mu_A}$ 39) most resembles b/w two objects
- 18) The missing element of PFS $(_, 0.2, 0.35)$ is 0.45
- 19) $(0.45, _, 0.55)$ missing elements of PFS 0
- 20) A fuzzy set $F_2 \{x, 0.2, 0.3\}$ is named as Hesitant fuzzy set
- 21) A hesitant fuzzy set is written [0, 1]
- 22) A hesitant fuzzy set $\forall x \in h(x)$ - lower bound $= h^-(x) = \min(h, x)$
- 23) A hesitant fuzzy set $\forall x \in h(x)$ $h^+(x) = \frac{1 - \mu}{1 - \mu^2}$
- 24) $h = \{0.6\}$ is HFS and $n = 2$ then $h^+ = \frac{1 - (1 - 0.6)^2}{1 - (1 - 0.6)^2} = 0.84$
- 25) $h(0.3)$ $n = 2$ $h^+ = 0.09$ 38) $\forall$ implication $x \vee y = \frac{(1 - x) \vee y}{1 - x}$
- 26) Hamming distance b/w set $h_1(x, 0.3)$ $h_2(x, 0.5) = 0.2$
- 27) Euclidean distance b/w $h_1(x, 0.4)$, $h_2(x, 0.7) = 0.09$
- 28) two object to be dissimilar if both have different shape
- 29) 1 implies 0 $\rightarrow 0$ 33) e^{x-1} $a \geq e^{-1}$
- 30) 0 implies 1 $\rightarrow 1$ 34) $R(x, y) = 1$, $R(y, z) = 1 \Rightarrow R(x, z)$
- 31) 1 implies 1 $\rightarrow 1$ 35) if $R(x) = e^{-\frac{1}{x}}$ $a \geq 0$
- 32) final e^{x-5} then $a = e^{-5}$ 36) Pick up instantaneous fuzzy set $\{x, (0.6, 0.3)\}$
- 37) b/w 0 then Property Nilpotent t-norm



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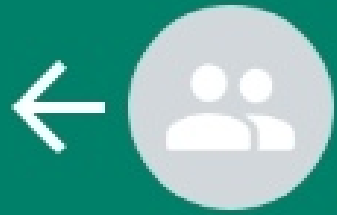
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